Agile **FitBuddy AI** project

1. Project Identification

Parameter	Details
Project Name	FitBuddy: AI Fitness Plan Generator Using Gemini Model
Team ID	GACSE-114
Submission Date	15 February 2026
Project Phase	Project Design Phase

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2. Agile Planning Logic (Sprint Breakdown)

The development is structured into two primary sprints to manage effort and feature delivery.

Sprint 1: Core Systems & AI Engine

Total Effort: 25 Story Points | User Story (USN) | Description | Effort (Points) | | :--- | :--- | :--- | | **USN1 - USN2** | Create input fields for Name, Age, Weight, and Height | 3 | | **USN3 - USN4** | Implement dropdowns for Fitness Goals and Diet Type selection | 4 | | **USN5 - USN6** | Backend logic for BMI and TDEE calculations | 6 | | **USN8 - USN9** | Groq API integration and AI fitness/diet plan generation | 10 |

Sprint 2: Visualization & Interaction

Total Effort: 23 Story Points | User Story (USN) | Description | Effort (Points) | | :--- | :--- | :--- | | **USN10 - USN12** | 30-day weight projection logic and line chart visualization | 7 | | **USN13 - USN15** | Streamlit chat interface with session state for context | 9 | | **USN16 - USN18** | UI layout optimization and AI prompt engineering | 7 |

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3. Solution Architecture Component Mapping

The following table describes the role of each technical component within the FitBuddy ecosystem.

Component	Technology	Responsibility
Frontend	Streamlit	Handles user profiles, goal setting, and plan display.
Backend	Python	Processes health metrics (BMI/TDEE) and data formatting.
AI Model	Llama 3.1 8B (Groq)	Generates personalized routines, diet plans, and advice.
Data Engine	NumPy / Pandas	Calculates 30-day weight projections and trends.
Memory	Session State	Maintains chatbot conversation history and context.

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4. Problem-Solution Fit Analysis

This table aligns the identified user pain points with the specific features provided by FitBuddy.

Customer Constraint (Problem)	FitBuddy Solution (Feature)
Information Overload	AI-driven filtering to provide one clear, personalized plan.
High Coaching Costs	Free access to professional-grade fitness and diet metrics.
Lack of Personalization	Plans generated based on unique BMI, TDEE, and diet goals.
Difficulty Tracking	30-day visual weight projections to maintain motivation.

Customer Constraint (Problem)	FitBuddy Solution (Feature)
Static Guidance	Interactive AI Chatbot for real-time follow-up questions.

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5. Technical Workflow (Data Flow)

1. **Capture:** User enters profile (Age, Weight, Height, Activity Level).
2. **Calculate:** Python backend computes **BMI** and **TDEE**.
3. **Prompt:** Processed data is sent as a structured prompt to the **Groq Llama 3.1** model.
4. **Visualize: NumPy** generates projection data displayed via Streamlit charts.
5. **Interact:** Users engage with the **AI Chatbot** for contextual advice using session state.